AI Based Face Recognition and Attendance Management System

Abstract

This project uses Machine Learning and Deep Learning to automate attendance using face recognition. It detects faces, extracts embeddings using CNN-based models, compares with enrolled faces, and marks attendance automatically.

Introduction

Traditional attendance is time-consuming and vulnerable to proxy attendance. AI-powered face recognition provides fast, contactless and accurate attendance.

Objectives

Develop an automatic attendance system; recognize registered students; reduce manual work; improve security; store attendance digitally.

Technologies

Python, OpenCV, TensorFlow/Keras or PyTorch, FaceNet/DeepFace, NumPy, Pandas, SQLite/MySQL.

Machine Learning

ML is used for classification and feature learning. Traditional methods include Eigenfaces, Fisherfaces and LBPH.

Deep Learning

CNN models learn facial features automatically. FaceNet generates embeddings for accurate face matching.

Methodology

Capture images, preprocess, detect face, extract features, compare with database, mark attendance, generate reports.

Dataset

Collect multiple images of each student under different lighting and angles; split into training and testing.

System Architecture

Camera -> Face Detection -> Feature Extraction -> Face Matching -> Attendance Database -> Report.

Advantages

Fast, accurate, contactless, prevents proxy attendance, saves time.

Limitations

Performance may reduce with poor lighting, masks, or low-quality cameras.

Applications

Schools, colleges, offices, laboratories, libraries and secure access systems.

Future Scope

Liveness detection, cloud deployment, mobile app, real-time analytics, multi-camera support.

Conclusion

AI-based attendance systems improve efficiency and accuracy using ML/DL techniques and are suitable for modern institutions.

References

OpenCV Documentation; FaceNet paper; TensorFlow Documentation; DeepFace GitHub.